

Thermalizing Quantum Systems & Evaluating Partition Functions with a Quantum Computer

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Definitions

- let H be a Hamiltonian with spectral decomposition

$$H = \sum_{a=1}^D E_a |\psi_a\rangle \langle \psi_a|, \quad E_a \in [0, 1]$$

- the partition function Z_β at inverse temperature β is

$$Z_\beta := \sum_{a=1}^D e^{-\beta E_a}$$

- the thermal state ρ_β at temperature β is

$$\rho_\beta := \sum_{a=1}^D \frac{e^{-\beta E_a}}{Z_\beta} |\psi_a\rangle \langle \psi_a|$$

Tasks

- Evaluating partition functions

given H , β , and ϵ , output a random value $\tilde{Z}_\beta$ such that

$$\Pr \left(|\tilde{Z}_\beta - Z_\beta| < \epsilon Z_\beta \right) > \frac{3}{4}$$

- Thermalizing quantum systems

given H , β , and ϵ , prepare a quantum state $\tilde{\rho}_\beta$ with

$$\frac{1}{2} \|\rho_\beta - \tilde{\rho}_\beta\|_{\text{tr}} \leq \epsilon$$

Thermalizing Quantum Systems

- given H , β , and ϵ , prepare a quantum state $\tilde{\rho}_\beta$ with

$$\frac{1}{2} \|\rho_\beta - \tilde{\rho}_\beta\|_{\text{tr}} \leq \epsilon \quad (*)$$

- the above task was studied by Terhal and Divincenzo in Phys. Rev. A 61, 022301
 - constructed a TCP whose fix-point is the thermal state ρ_β in an idealized setting (perturbation theory)
 - did not prove how the number of iterations has to be chosen to obtain a state $\tilde{\rho}_\beta$ satisfying (*)
 - numerical simulations suggests that this algorithm may outperform classical algorithms

Our Quantum Algorithm

- given H , β , and ϵ , our quantum algorithm makes it possible to prepare $\tilde{\rho}_\beta$ with

$$\frac{1}{2} \|\rho_\beta - \tilde{\rho}_\beta\|_{\text{tr}} \leq \epsilon$$

- its running time is

$$\sqrt{\frac{D}{Z_\beta}} \cdot \text{poly}(\log D, 1/\epsilon, \beta)$$

Classical Hamiltonians

- let Ω be a set whose elements x correspond the states of some classical system
- let $E : \Omega \rightarrow [0, 1]$ denote the energy function, assigning to each state x its energy $E(x)$
- let

$$H = \sum_{x \in \Omega} E_x |x\rangle \langle x|$$

- thermalizing such classical systems is easier because H is diagonal in the computational basis $\{|x\rangle\}_{x \in \Omega}$

Metropolis Algorithm

- given such classical Hamiltonian H , we can always construct a Markov chain P whose limiting distribution is equal to ρ_β
- the spectral gap δ determines how fast we can approach the Boltzmann distribution
- the run time scales like $1/\delta$
- often, it is not possible to determine (or bound) the spectral gap $\Rightarrow$ heuristics

Quantized Metropolis Algorithm

- given such classical Hamiltonian H , there is a “quantized” Metropolis algorithm preparing $|\tilde{\varphi}_\beta\rangle$ with

$$\| |\varphi_\beta\rangle - |\tilde{\varphi}_\beta\rangle \| \leq \epsilon$$

where $|\varphi_\beta\rangle$ is a “quantized” version of the Boltzmann distribution

$$|\varphi_\beta\rangle := \sum_{x \in \Omega} \sqrt{\frac{e^{-\beta E_x}}{Z_\beta}} |x\rangle$$

- the run times scales like $\sqrt{1/\delta}$
- proved in Somma, Boixo, Barnum, and Knill, PRL 101, 130504 and W. and Abeyesinghe, PRA 78, 032311

Grover's State Engineering

- prepare the state

$$\frac{1}{\sqrt{D}} \sum_{x \in \Omega} |x\rangle \otimes |E_x\rangle \otimes |0\rangle$$

- apply a controlled rotation to obtain the state

$$\frac{1}{\sqrt{D}} \sum_{x \in \Omega} |x\rangle \otimes |E_x\rangle \otimes \left(\sqrt{e^{-\beta E_x}} |0\rangle + \sqrt{1 - e^{-\beta E_x}} |1\rangle \right)$$

Grover's State Engineering

- apply Grover to increase the overlap with the $|0\rangle$ component and measure in the basis $|0\rangle, |1\rangle$
- if we obtain $|0\rangle$, then the post-measurement state is

$$\sum_{x \in \Omega} \sqrt{\frac{e^{-\beta E_x}}{Z_\beta}} |x\rangle \otimes |E_x\rangle \otimes |0\rangle$$

- the expected run time to obtain this state scales like

$$\sqrt{\frac{D}{Z_\beta}}$$

Our Quantum Algorithm

- is a generalization of the algorithm based on Grover's state engineering to **quantum** Hamiltonians
- let H be an arbitrary Hamiltonian with spectral decomposition

$$H = \sum_{a=1}^D E_a |\psi_a\rangle \langle \psi_a|, \quad E_a \in [0, 1]$$

Error Sources

- given H , β , and ϵ , our quantum algorithm prepares $\tilde{\rho}_\beta$

$$\frac{1}{2} \|\rho_\beta - \tilde{\rho}_\beta\|_{\text{tr}} \leq \epsilon$$

- the errors come from three sources:

$$\|\rho_\beta - \tilde{\rho}_\beta\|_{\text{tr}} \leq \|\rho - \rho_{\text{sim}}\|_{\text{tr}} + \|\rho_{\text{sim}} - \rho_{\text{prec}}\|_{\text{tr}} + \|\rho_{\text{prec}} - \rho_{\text{fail}}\|_{\text{tr}}$$

- ρ_{sim} due to imperfect simulation of $\exp(iH)$
- ρ_{prec} due to limited precision of phase estimation (PE)
- $\tilde{\rho}_\beta = \rho_{\text{fail}}$ due to nonzero failure probability of PE

Imperfect Hamiltonian Simulation

- assume that H is sparse and let $U = \exp(iH)$
- using techniques for simulating Hamiltonian time evolutions, we can implement U_{sim} with

$$\|U - U_{\text{sim}}\| \leq \epsilon_{\text{sim}}$$

- the run time scales like

$$\text{poly}(\log D, 1/\epsilon_{\text{sim}}, s)$$

where s is the number of non-zero entries maximized over all rows

Technical Arguments

- matrix logarithm is Lipschitz continuous:

$$\|U - U_{\text{sim}}\| \leq \epsilon_{\text{sim}}$$

implies that there is a Hamiltonian H_{sim} with

$$\|H - H_{\text{sim}}\| \leq O(\epsilon_{\text{sim}})$$

- thermal states are Hölder continuous:

$$\|H - H_{\text{sim}}\| \leq O(\epsilon_{\text{sim}})$$

implies that the corresponding thermal states satisfy

$$\|\rho - \rho_{\text{sim}}\|_{\text{tr}} \leq O(\sqrt{\beta \epsilon_{\text{sim}}})$$

- we assume $U_{\text{sim}} = U$ (and keep track of the error)

Maximally Entangled State

• let

$$|\Phi\rangle := \frac{1}{\sqrt{D}} \sum_{a=1}^D |i\rangle \otimes |i\rangle$$

• let

$$V = \sum_{a=1}^D |\psi_a\rangle \langle a|$$

• it is important that we can express $|\Phi\rangle$ as

$$|\Phi\rangle = \sum_{a=1}^D |\psi_a\rangle \otimes |\psi_a^T\rangle$$

this follows from $(V \otimes I_D)|\Phi\rangle = (I_D \otimes V^T)|\Phi\rangle$

Phase Estimation with Boosting

● recall that $U = \exp(iH)$

● $\Rightarrow$ we have

$$(U \otimes I_D)|\psi_a\rangle \otimes |\psi_a^T\rangle = e^{iE_a}|\psi_a\rangle \otimes |\psi_a^T\rangle$$

$$(U \otimes I_D)|\Phi\rangle = \sum_{a=1}^D e^{iE_a}|\psi_a\rangle \otimes |\psi_a^T\rangle$$

● $\Rightarrow$ the energy levels E_a of H are mapped bijectively onto the phases e^{iE_a} of U since no wrapping around

● $\Rightarrow$ apply PE to estimate the energy levels

Phase Estimation with Boosting

- use PE with $O(1/\epsilon_{\text{prec}})$ ancillas and run $O(\log(1/\epsilon_{\text{fail}}))$ boosting rounds
 $\Rightarrow O((1/\epsilon_{\text{prec}}) \log(1/\epsilon_{\text{fail}}))$ applications of controlled- U
- PE with boosting transforms $|\Phi\rangle$ into the state $|\tilde{\Phi}_\beta\rangle$

$$\sum_{a=1}^D |\psi_a\rangle \otimes |\psi_a^T\rangle \otimes \left(c_a^- |E_a^-\rangle + c_a^+ |E_a^+\rangle \right) + |\xi\rangle$$

$$\bullet E_a^+ - E_a \leq \epsilon_{\text{prec}}, \quad E_a - E_a^- \leq \epsilon_{\text{prec}}$$

$$\bullet |c_a^+|^2 + |c_a^-|^2 = 1$$

$$\bullet |||\xi\rangle|| \leq \epsilon_{\text{fail}}$$

Grover's State Engineering

- adjoin to $|\tilde{\Phi}_\beta\rangle$ a qubit prepared in $|0\rangle$ and apply a rotation controlled by the value in the energy register
- this yields the state $|\tilde{\Psi}_\beta\rangle$

$$\sum_{a=1}^D |\psi_a\rangle \otimes |\psi_a^T\rangle \otimes c_a^\pm |E_a^\pm\rangle \otimes \left(\sqrt{e^{-\beta E_a^\pm}} |0\rangle + \sqrt{1 - e^{-\beta E_a^\pm}} |1\rangle \right) + |\tilde{\xi}\rangle$$

Outline Analysis

- if perfect PE were possible, we could prepare the state

$$|\Psi_\beta\rangle = \sum_{a=1}^D |\psi_a\rangle \otimes |\psi_a^T\rangle \otimes |E_a\rangle \otimes \left(\sqrt{e^{-\beta E_a}} |0\rangle + \sqrt{1 - e^{-\beta E_a}} |1\rangle \right)$$

- $\Rightarrow$ using Grover as in the case of classical Hamiltonians, we could obtain a purification of the thermal state

$$|\varphi\rangle = \sum_{a=1}^D \sqrt{\frac{e^{-\beta E_a}}{Z_\beta}} |\psi_a\rangle \otimes |\psi_a^T\rangle \otimes |E_a\rangle \otimes |0\rangle$$

- the desired thermal state would be obtained by tracing out the last three registers

Outline of Analysis

- a careful analysis shows that the thermal state $\tilde{\rho}_\beta$ obtained by first applying Grover to the imperfect state $|\tilde{\Psi}_\beta\rangle$ and then tracing out is sufficiently close to ρ_β
- in summary, given H , β , and ϵ , our quantum algorithm makes it possible to prepare $\tilde{\rho}_\beta$

$$\frac{1}{2} \|\rho_\beta - \tilde{\rho}_\beta\|_{\text{tr}} \leq \epsilon$$

- the expected run time is

$$\sqrt{\frac{D}{Z_\beta}} \cdot \text{poly}(\log D, 1/\epsilon, \beta)$$

Summary & Future Research

- quantum algorithm making it possible to approximately prepare thermal states
- rigorous proof of running time in terms of Z_β , β , and ϵ
- Open question: Is there a truly quantum analog of the Metropolis algorithm?

(not to be confused with the quantized Metropolis algorithm, which works only for classical Hamiltonians)